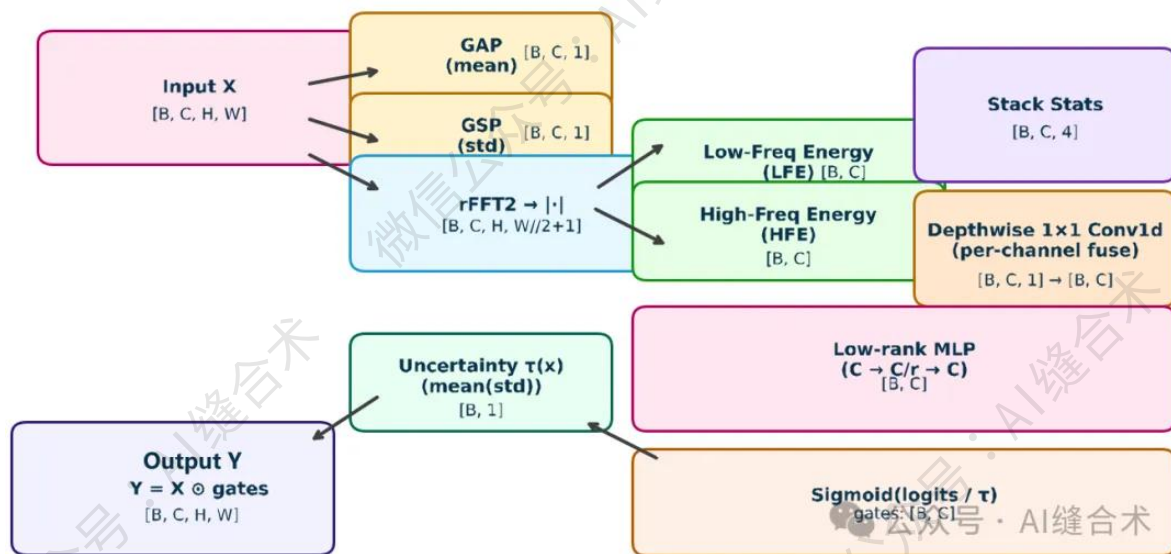


核心速览



```
2025 > 120.0 (2025 AI缝合术原创) FU-SE.py
1 import torch
2 import torch.nn as nn
3 import torch.nn.functional as F
4
5 class FUSEChannelAttention(nn.Module):
6     def __init__(self, channels, reduction=8, lf_ratio=0.125, eps=1e-6):
7         super().__init__()
8         self.C = channels
9         self.r = max(1, reduction)
10        self.lf_ratio = lf_ratio
11        self.eps = eps
12
13        # 用可学习的 per-channel 权重 (C x 4) 和 bias (C,)
14        self.per_channel_weight = nn.Parameter(torch.full((channels, 4), 0.25))
15        self.per_channel_bias = nn.Parameter(torch.zeros(channels))
16
17        # 低秩跨通道混合 (类似 SE 的 MLP)
18        hidden = max(1, channels // self.r)
19        self.mlp = nn.Sequential(
20            nn.Linear(channels, hidden, bias=True),
21            nn.ReLU(inplace=True),
22            nn.Linear(hidden, channels, bias=True)
23        )
24
25    def forward(self, x):
26        # GAP (mean)
27        gap = torch.mean(x, dim=[2, 3], keepdim=True)
28        # GSP (std)
29        gsp = torch.std(x, dim=[2, 3], keepdim=True)
30        # rFFT2 → |·|
31        rfft2 = torch.fft.rfft2(x, dim=[2, 3])
32        mag = torch.abs(rfft2)
33        # Low-Freq Energy (LFE)
34        lf_ratio = self.lf_ratio
35        lf = mag[:, :, :lf_ratio * mag.shape[2], :].sum(dim=[2, 3], keepdim=True)
36        # High-Freq Energy (HFE)
37        hfe = mag[:, :, lf_ratio * mag.shape[2]:, :].sum(dim=[2, 3], keepdim=True)
38        # Stack Stats
39        stats = torch.cat([gap, gsp, lf, hfe], dim=-1)
40        # Depthwise 1x1 Conv1d (per-channel fuse)
41        conv1d = torch.nn.Conv1d(stats.shape[-1], self.C, kernel_size=1, bias=False)
42        conv1d.weight = nn.Parameter(self.per_channel_weight)
43        conv1d.bias = nn.Parameter(self.per_channel_bias)
44        conv1d_stats = conv1d(stats)
45        # Low-rank MLP (C → C/r → C)
46        mlp = self.mlp(conv1d_stats)
47        # Uncertainty τ(x) (mean(std))
48        uncertainty = torch.mean(gsp, dim=-1, keepdim=True)
49        # Sigmoid(logits / τ)
50        logits = mlp + uncertainty
51        gates = torch.sigmoid(logits)
52        # Output Y
53        Y = x * gates
54        return Y
```

微信公众号: AI缝合术

输入张量形状: torch.Size([1, 32, 64, 64])
输出张量形状: torch.Size([1, 32, 64, 64])

用 GPT-5 生成新型通道注意力 Fuse-SE, 代码 0 错误!

论文内容详细解读: <https://mp.weixin.qq.com/s/adS3rsOB2vqxqvXlf8zABA>